Code No: 135SC **RA**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, October - 2020
OPERATING SYSTEMS
(Common to CSE, IT)

Time: 2 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Define operating system. Elaborate the operating system operations with examples.
b) Briefly explain typical functions of an operating system kernel. [7+8]

2. Construct the grant chart for (a) shortest job first (b) Round robin with q=3 (c) Round robin with q=4 (d) Shortest remaining time first scheduling algorithms for the following. [15]

Process ID	Arrival Time (in ms)	CPU Burst Time (in ms)
1	0	4
2	1	3
3	2	1
4	3	2
5	4	5

3.a) Explain about FIFO, LRU page replacement algorithms with example.
b) Differentiate internal fragmentation and external fragmentation. [7+8]

4.a) Explain the various methods of file sharing.
b) What is file structure? How file structure is supported by different operating systems? [7+8]

5. Explain the Banker's algorithm for deadlock avoidance. [15]

6.a) Define and explain various types of operating systems.
b) Identify three advantages and one disadvantage of multiprocessor systems. [7+8]

7. Consider the following set of process that arrive at time 0, with the length of the CPU burst given in milliseconds.

Process	Burst time
P1	2
P2	3
P3	3

Calculate the average waiting time when the processes arrive in the following order a) p1, p2, p3 b) p2, p3, p1 provide the grant chart for the same using FCFS, SJF scheduling algorithms. [15]

8.a) Define thrashing? Explain its causes and write any two solutions to increase CPU Utilization in case of thrashing.
b) Explain about segmented memory management with a neat block diagram. [8+7]

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